

## Exp no:10

**Implement the Boundary condition between two different dielectric media in Magnetic Field and find the relative permeability of the different medium.**

### Aim:

To write program for Boundary condition between two different dielectric media in magnetic field using SCILAB.

### Software required:

- SCILAB version 6.0.1

```
Scilab 2026.1.0 Console
File Edit Control Applications ?
Enter the value of theta1=50
Enter the value of theta2=80
Enter the value of m1=4
"THE VALUE OF M2 IS"
19.035082
-->

Exp No 10.sci (D:\HS-Sem Courses\Engineering Electromagnetic...
File Edit Format Options Window Execute ?
Exp No 10.sci (D:\HS-Sem Courses\Engineering Electromagnetic lab exp\Ex9)
Exp No 2.sci Exp No 3 case 1.sci Exp No 3.sci Exp No 10.sci
1 |clc;
2 |clear;
3 |theta_1=input("Enter the value of theta1=");
4 |theta_2=input("Enter the value of theta2=");
5 |m_1=input("Enter the value of m1=");
6 |tan_theta= tand(theta_1)/tand(theta_2);
7 |m_2= m_1/tan_theta;
8 |disp('THE VALUE OF M2 IS', [m_2]);
9
10

History
03
004
10^-6
10^-6
03
004
10/08/2026 11:37:44 --//
Discover Scilab community
Scilab is open source.
Giving back support or use cases is appreciated.
Add and improve functions on Scilab.
View and edit help pages.
Publish toolboxes on ATOMS.
Post questions on Discourse.
```

EXP NO: -10

$$\theta_1 = 50^\circ$$

$$\theta_2 = 80^\circ$$

$$\mu_2 = \mu_1 \frac{\tan \theta_2}{\tan \theta_1}$$

$$\tan 50^\circ = 1.1918$$

$$\tan 80^\circ = 5.6713$$

$$\frac{\tan 80^\circ}{\tan 50^\circ} = \frac{5.6713}{1.1918} = 4.7587$$

$$\mu_2 = 1 \times 4.7587 = 4.7587$$

$$\mu_2 = 19.035$$

### Result :

Thus the program was verified for Boundary condition between two different dielectric media in magnetic field using SCILAB was successfully.

**Case 1:** Consider two identical media in the presence of a magnetic field. The relative permeability in both media is 1. What will be the effect of this on the magnetic field and flux density across the boundary?

The screenshot shows the Scilab 2026.1.0 environment. The console window on the left displays the execution of a script, showing the calculated values for magnetic field intensity and flux density in two identical media. The editor window on the right shows the source code for 'Exp No 10 case 1.sci'.

```

// Scilab code to simulate the magnetic field and flux density for two identical media
// Constants
mu_0=4*pi*10^(-7); // Permeability of free space (H/m)
H1=10; // Magnetic field intensity in medium 1 (A/m)
// Relative permeability for both media (both are 1)
mu_r1=1; // Relative permeability for medium 1
mu_r2=1; // Relative permeability for medium 2
// Calculate the magnetic field intensity in both media (no change because mu_r = 1)
H2=H1; // Since the media are identical, H1 = H2
// Magnetic flux density in both media (B = mu * H)
B1=mu_0*mu_r1*H1; // Magnetic flux density in medium 1
B2=mu_0*mu_r2*H2; // Magnetic flux density in medium 2
// Display results
disp("Magnetic Field Intensity in Medium 1: "+string(H1)+" A/m");
disp("Magnetic Field Intensity in Medium 2: "+string(H2)+" A/m");
disp("Magnetic Flux Density in Medium 1: "+string(B1)+" T");
disp("Magnetic Flux Density in Medium 2: "+string(B2)+" T");

```

Console Output:

```

Enter the value of theta1=50
Enter the value of theta2=80
Enter the value of m1=4
"THE VALUE OF M2 IS"
19.035082
--> exec('D:\HS-Sem Courses\Engineering Electromagnetic\Engineering Electromagnetic lab 2\Exp No 10 case 1.sci')
"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 10 A/m"
"Magnetic Flux Density in Medium 1: 0.0000126 T"
"Magnetic Flux Density in Medium 2: 0.0000126 T"
-->

```

**Result :** Thus, the effect of two identical media on the magnetic field and flux density across the boundary is calculated.

**Case 2:** In a system with two media, where the relative permeability of medium 1 is  $\mu_{r1}=4$  and medium 2 is a vacuum ( $\mu_{r2} = 1$ ), how would the magnetic field intensity and flux density behave at the boundary?

The screenshot shows the Scilab 2026.1.0 interface. The console window on the left displays the execution of a script that calculates magnetic field intensity and flux density in two media. The editor window on the right shows the source code for 'Exp No 10 case 2.sci', which simulates the magnetic field intensity and flux density across a boundary.

```

--> exec('D:\HS-Sem Courses\Engineering Electromagnetic\Engineering Electromagnetic lab\Exp No 10 case 2.sci')
"Magetic Field Intensity in Medium 1: 10 A/m"
"Magetic Field Intensity in Medium 2: 40 A/m"
"Magetic Flux Density in Medium 1: 0.0000503 T"
"Magetic Flux Density in Medium 2: 0.0000503 T"

-->

1 // Scilab code to simulate the magnetic field intensity and flux density across the boundary
2
3 // Constants
4 mu_0=4*pi*10^(-7); // Permeability of free space (H/m)
5 H1=10; // Magnetic field intensity in medium 1 (A/m)
6
7 // Relative permeability for the media
8 mu_r1=4; // Relative permeability for medium 1
9 mu_r2=1; // Relative permeability for medium 2 (vacuum)
10
11 // Calculate the magnetic field intensity in medium 2 using the boundary condition
12 H2=mu_r1*H1/mu_r2; // H2 = (mu_r1 / mu_r2) * H1
13
14 // Calculate the magnetic flux density in both media (B = mu * H)
15 B1=mu_0*mu_r1*H1; // Magnetic flux density in medium 1
16 B2=mu_0*mu_r2*H2; // Magnetic flux density in medium 2
17
18 // Display results
19 disp("Magnetic Field Intensity in Medium 1: "+string(H1)+" A/m");
20 disp("Magnetic Field Intensity in Medium 2: "+string(H2)+" A/m");
21
22 disp("Magnetic Flux Density in Medium 1: "+string(B1)+" T");
23 disp("Magnetic Flux Density in Medium 2: "+string(B2)+" T");
24

```

## Result :

Thus, the effect of two different media on the magnetic field and flux density across the boundary is calculated.

**Case 3:** If medium 1 has a relative permeability  $\mu_{r1}=0.3$  and medium 2 has a relative permeability  $\mu_{r2}=2$ , calculate the magnetic field intensity in both media at the boundary if the magnetic field intensity in medium 1 is known to be  $H_1=10$  A/m.

The screenshot displays the Scilab 2026.1.0 environment. The console window on the left shows the execution of a script with the following output:

```
"Magnetic Field Intensity in Medium 1: 10 A/m"
"Magnetic Field Intensity in Medium 2: 1.5 A/m"
"Magnetic Flux Density in Medium 1: 0.0000038 T"
"Magnetic Flux Density in Medium 2: 0.0000038 T"
exec: Wrong number of output argument(s): 0 expected.
```

The editor window on the right shows the script being executed, titled "Exp No 10 case 3.sci". The script calculates the magnetic field intensity and flux density in two media, Medium 1 and Medium 2, based on the input values provided in the console output.

```
1 clear;
2 clear;
3
4 // Relative permeability
5 mu_r1 = 0.3;
6 mu_r2 = 2;
7
8 // Permeability of free space
9 mu_0 = 4 * pi * 1e-7;
10
11 // Magnetic field intensity in Medium 1
12 H1 = 10;
13
14 // Calculate magnetic field intensity in Medium 2
15 H2 = (mu_r1 / mu_r2) * H1;
16
17 // Calculate magnetic flux density
18 B1 = mu_0 * mu_r1 * H1;
19 B2 = mu_0 * mu_r2 * H2;
20
21 // Display results
22 disp("Magnetic Field Intensity in Medium 1: " + string(H1) + " A/m");
23 disp("Magnetic Field Intensity in Medium 2: " + string(H2) + " A/m");
24
25 disp("Magnetic Flux Density in Medium 1: " + string(B1) + " T");
26 disp("Magnetic Flux Density in Medium 2: " + string(B2) + " T");
27
```

## Result:

Thus, the effect of two different media on the magnetic field and flux density across the boundary is calculated.